THRESHOLD ENTERPRISE SYSTEMS
AI Model Deployment Policy

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1. RESTRICTED CLASSIFICATION AND RIGOROUS OPERATIONAL PURPOSE

*** RESTRICTED DOCUMENT - STRICTLY CONFIDENTIAL TO AUTHORIZED TECHNICAL PERSONNEL ***

THIS DOCUMENT CONTAINS HIGHLY SENSITIVE PRODUCTION DEPLOYMENT PROTOCOLS, CRYPTOGRAPHIC SIGN-OFF PROCEDURES, INFRASTRUCTURE CANARY GATES, AND AIR-GAPPED ROLLBACK ARCHITECTURES FOR THRESHOLD ENTERPRISE SYSTEMS.

CLASSIFICATION: RESTRICTED.

AUTHORIZED ACCESS ROLES: Strictly limited to AI_ENGINEER, SECURITY_ENGINEER, and ADMIN. All other enterprise roles—including EMPLOYEE, MANAGER, ENGINEER, OPERATIONS_MANAGER, and HR_MANAGER—are categorically denied access to this document within the enterprise knowledge base and document registry.

The operational purpose of this AI Model Deployment Policy is to establish an uncompromising, multi-gated production release framework that ensures no artificial intelligence model, RAG system, or autonomous agent reaches live enterprise production without exhaustive technical verification, multi-party cryptographic authorization, and guaranteed fail-safe rollback capabilities.

2. Pre-Deployment Governance Prerequisites: AIG-002 and AIG-003 Dependencies

Deployment authorization is fundamentally conditional and cannot be initiated without verified completion of prior governance milestones:

1. Certified Risk Classification (AIG-002 Dependency): The candidate model must possess a verified, unexpired Risk Tier Assignment (LOW, MEDIUM, HIGH, or CRITICAL) cataloged in the Enterprise AI Registry.

2. Formal Model Approval Sign-Off (AIG-003 Dependency): The candidate model must hold an active, unconditional Model Approval Certification issued by the appropriate governing authority.

3. Mandatory Dossier Completeness: The Model Card, Data Sheet, Bias Audit Report, and Human Oversight Runbook must be cryptographically signed and archived in the governance repository.

Attempting to bypass pre-deployment governance dependencies or deploy an unapproved model via automated CI/CD script overrides constitutes a catastrophic security violation resulting in immediate privilege revocation and termination.

3. The Seven Technical Pre-Flight Verification Gates

Before production promotion scripts can be executed, the deployment pipeline automatically validates seven mandatory pre-flight technical gates:

Gate 1 - Container Vulnerability Scanning: Model serving container images must pass automated vulnerability scans with zero Critical or High CVEs and 100% compliant software bill of materials (SBOM).

Gate 2 - Weight Integrity and Cryptographic Attestation: Production model weights and configuration manifests must match SHA-256 cryptographic signatures registered during AIG-003 approval.

Gate 3 - Latency and Throughput Stress Testing: The candidate model must satisfy P95 and P99 latency budgets under 150% simulated peak production concurrency in staging.

Gate 4 - Privacy Sanitization Verification (AIG-006 Gate): Inference logging pipelines must demonstrate active PII redaction and vector partitions must verify tenant metadata isolation.

Gate 5 - Adversarial Penetration Testing (AIG-007 Gate): The staging endpoint must withstand 100% of standard prompt injection regression test payloads without guardrail bypass.

Gate 6 - Monitoring Telemetry Instrumentation (AIG-008 Gate): Real-time metrics streaming, drift calculation workers, and PagerDuty alert hooks must be verified and operational.

Gate 7 - Human Oversight Interface Validation (AIG-005 Gate): Review queues, override actions, and justification logging mechanisms must be tested and verified by designated human reviewers.

4. Staged Canary Deployment and Soak Schedule

To prevent systemic blast radius in the event of unforeseen edge-case failure, all production deployments must adhere to a strict, progressive canary traffic rollout schedule:

- Canary Stage 1 (Internal Shadow Mode): The model processes 100% of production traffic in shadow mode for a minimum of forty-eight (48) hours. Outputs are logged and compared against baseline models; zero live user responses are served.

- Canary Stage 2 (5% Live Traffic Rollout): The model serves 5% of live traffic to randomized user cohorts for twenty-four (24) hours. Latency, error rates, and hallucination indicators are scrutinized continuously.

- Canary Stage 3 (25% Live Traffic Expansion): Traffic expands to 25% for twenty-four (24) hours, validating database connection pooling, vector index caching, and inference cluster autoscaling.

- Canary Stage 4 (50% Live Traffic Balanced): Traffic expands to 50% for twelve (12) hours with balanced blue/green cluster monitoring.

- Canary Stage 5 (100% Full Production Promotion): Following unanimous verification of stability, traffic routes 100% to the new release. The previous stable version remains warm on standby for seventy-two (72) hours.

5. Automated Rollback Tripwires and Instant Rollback Procedures

During canary rollout and active production operation, automated tripwires trigger instantaneous rollback if any of the following conditions occur:

- Latency Breach: P99 inference latency exceeds the SLA budget (600ms) by more than 30% over any three-minute rolling window.

- Error Spike: HTTP 5xx error responses exceed 1.5% of total request volume over a two-minute window.

- Guardrail Failure: The prompt firewall detects any bypass of core safety guardrails or canary token leakage.

- Hallucination Threshold Breach: Real-time sampling indicates a hallucination rate exceeding 4.0%.

Rollback Execution: The automated CI/CD orchestrator switches DNS and load balancer targets back to the standby baseline container within 15 seconds, immediately cutting traffic from the candidate release and preserving audit logs for post-incident triage.

6. Blue/Green Cluster Orchestration and Zero-Downtime Deployment

Production AI model releases must leverage blue/green infrastructure clusters to guarantee continuous availability without service interruption:

- Parallel Green Cluster Provisioning: The candidate model is spun up on an isolated, identical production infrastructure cluster (the Green environment) alongside the active production cluster (the Blue environment).

- Synthetic Smoke Testing: Automated smoke tests simulate authenticated inference queries against the Green cluster, verifying vector database connectivity, GPU memory allocation, and token generation speed.

- Weighted DNS Traffic Shifting: Application load balancers gradually shift traffic weights from Blue to Green according to the canary soak schedule.

- Standby Blue Cluster Warm Retention: The Blue cluster remains powered on and warm for seventy-two (72) hours post-cutover. If an undetected defect surfaces, traffic is instantly reverted to Blue via a single load balancer weight switch.

7. Secrets Management, Environment Isolation, and Runtime Security

Production model serving clusters operate under strict zero-trust infrastructure parameters:

- Zero Hardcoded Secrets: All API tokens, vector database credentials, and cryptographic keys must be dynamically injected into runtime memory using HashiCorp Vault with 60-minute automatic rotation.

- Egress Network Filtering: Inference pods are restricted by network security policies, permitting outbound connections exclusively to certified vector database endpoints and logging daemons. Arbitrary public internet egress is blocked.

- Ephemeral Scratch Storage: Temporary inference scratch files, model caches, and intermediate tensors must reside in encrypted RAM-disks (tmpfs) that are wiped automatically upon container termination.

8. Emergency Rollback Drills and Disaster Recovery Failover

To guarantee operational resilience during severe infrastructure outages or model failures, engineering squads must execute routine rollback validation drills:

- Automated Staging Rollback Exercises: Prior to production promotion, the CI/CD pipeline executes an automated simulated rollback test in staging, validating that DNS failover and traffic rerouting complete in under fifteen (15) seconds without dropped TCP connections.

- Multi-Region Standby Redundancy: For CRITICAL risk models, cold-standby infrastructure clusters are provisioned in a secondary geographic cloud region (e.g., US-East secondary for US-West primary), with synchronized vector stores and model weights ready for immediate failover.

- Disaster Recovery Objectives: Production AI services must maintain a Recovery Time Objective (RTO) of less than five (5) minutes and a Recovery Point Objective (RPO) of zero data loss for transactional vector indices.

9. Dual-Custody Production Authorization Protocol

Promotion of any HIGH or CRITICAL risk AI model to 100% production traffic requires dual-custody cryptographic sign-off (the Two-Person Rule):

- Signer 1: Lead AI Engineer certifying technical benchmark compliance, latency stability, and functional correctness.

- Signer 2: Information Security Engineer certifying adversarial robustness, container security, and vulnerability remediation.

Both signatures must be executed using enterprise hardware security tokens (YubiKey) and recorded immutably in the production release ledger before deployment locks are released.

10. Post-Deployment Verification and 72-Hour Hypercare Period

Upon achieving 100% traffic promotion, the engineering squad enters a mandatory 72-hour Hypercare Period:

- Dedicated On-Call Shift: A designated AI Engineer and Security Engineer maintain primary on-call readiness with a 15-minute response SLA.

- Daily Stakeholder Briefings: Daily 15-minute operational check-ins review user feedback, drift indicators, and reviewer override logs.

- Formal Release Closure: At the conclusion of 72 hours with zero Severity 1 or 2 incidents, the Hypercare Period terminates, the deployment status is updated to ACTIVE in the Enterprise AI Registry, and standby legacy nodes are safely decommissioned.

11. Audit Records, Governance Compliance, and References

All deployment artifacts—including container digests, git commit SHAs, cryptographic attestation signatures, canary telemetry charts, and rollback event logs—are archived in the Enterprise Governance Registry for a minimum of seven (7) years.

This policy operates in direct alignment with related enterprise governance standards:

- AIG-002: AI Risk Classification Framework (Sets risk tiers governing canary duration).

- AIG-003: AI Model Approval Policy (Mandates prior approval certificate).

- AIG-005: Human-in-the-Loop Review Policy (Validates HITL interface readiness).

- AIG-006: AI Data Privacy Policy (Validates PII scrubbing and data boundaries).

- AIG-008: AI Model Monitoring Policy (Verifies telemetry streaming and alert hooks).